

Tutorial 3: Requirements Analysis – Dynamic Models

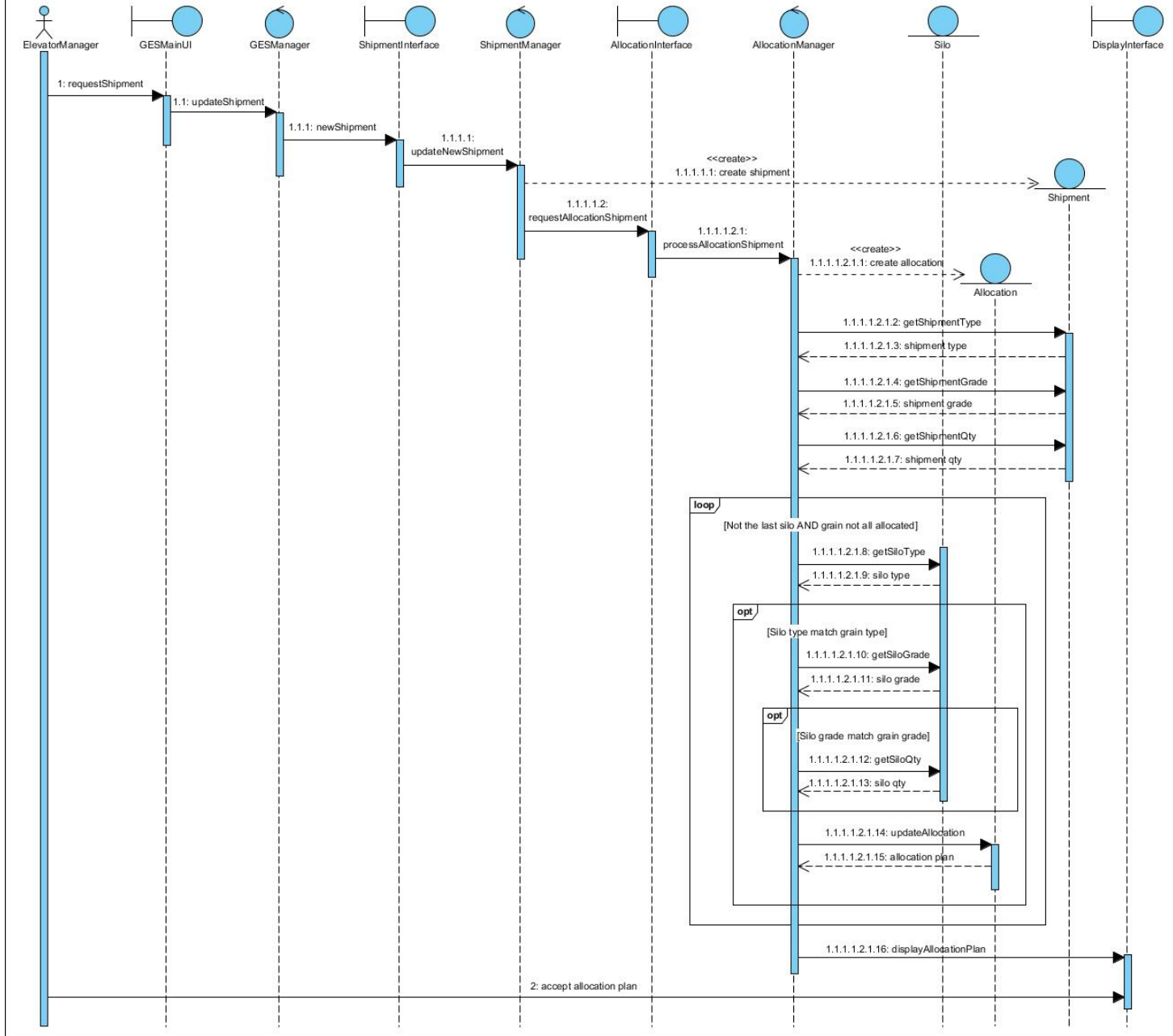
Question 1: Grain Elevator System (GES)

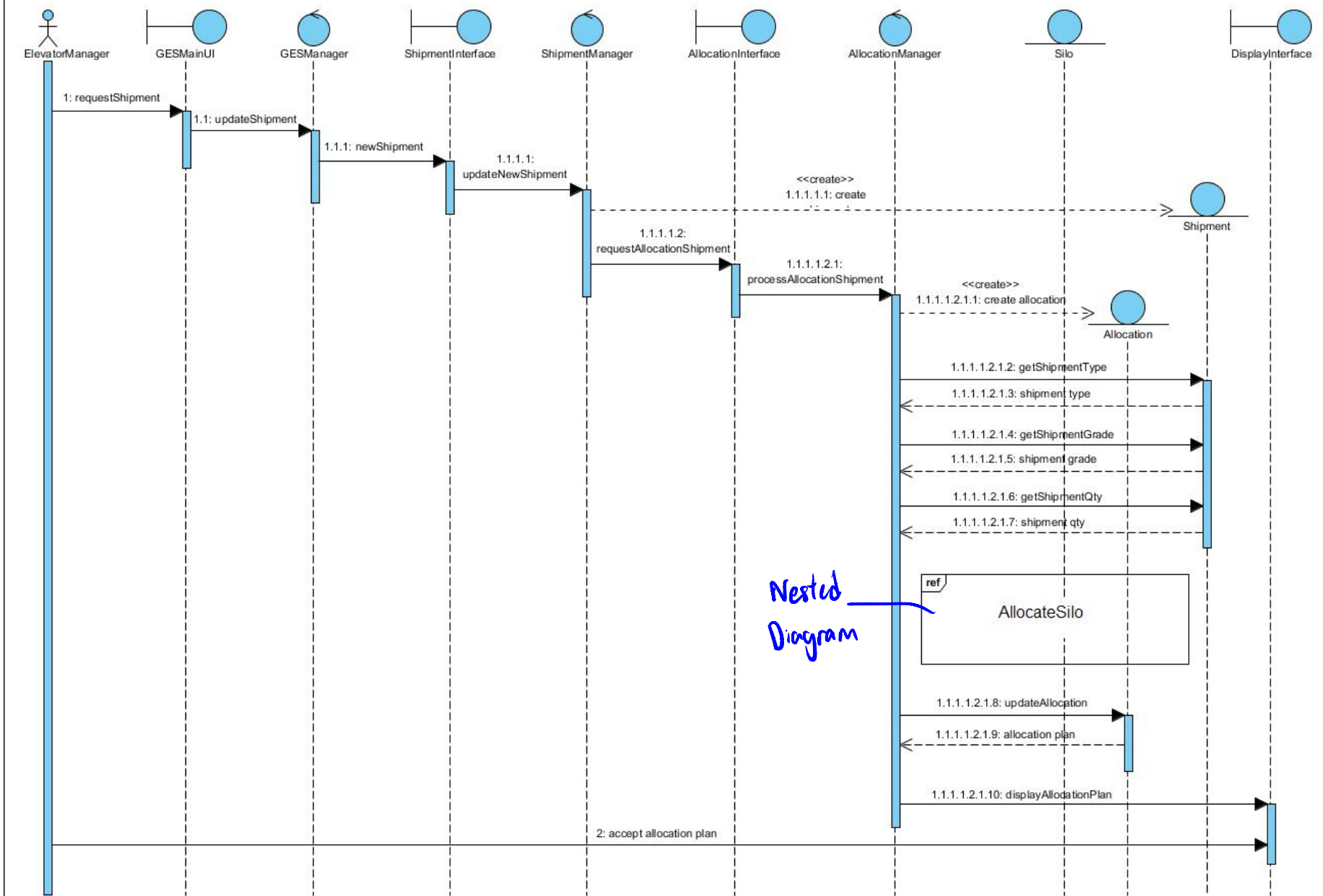
- (a) Draw a sequence diagram to model the interaction amongst the classes to enact the shipment arrival functionality.
- (b) Refine the conceptual model from last week tutorial.

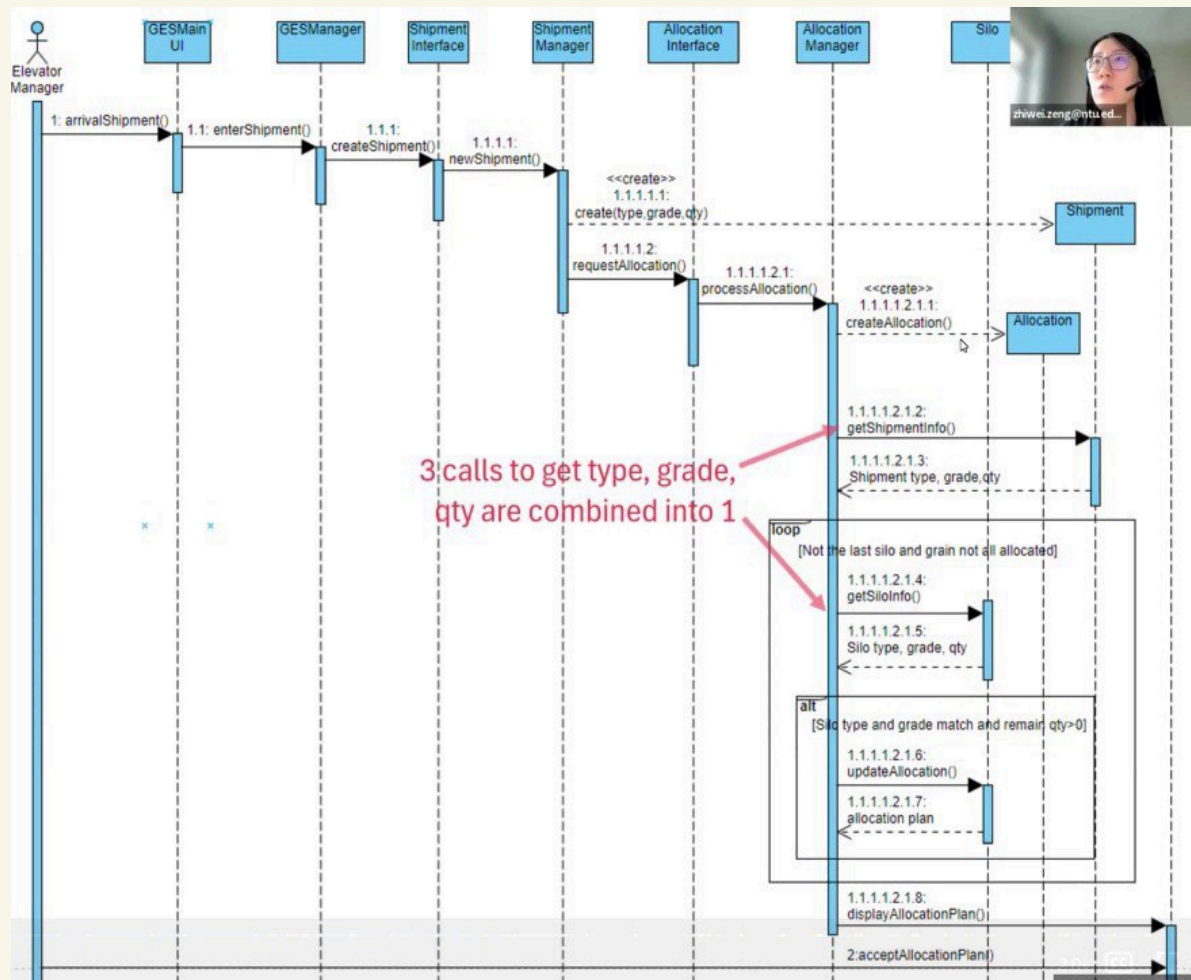
From Use Case to Sequence Diagram



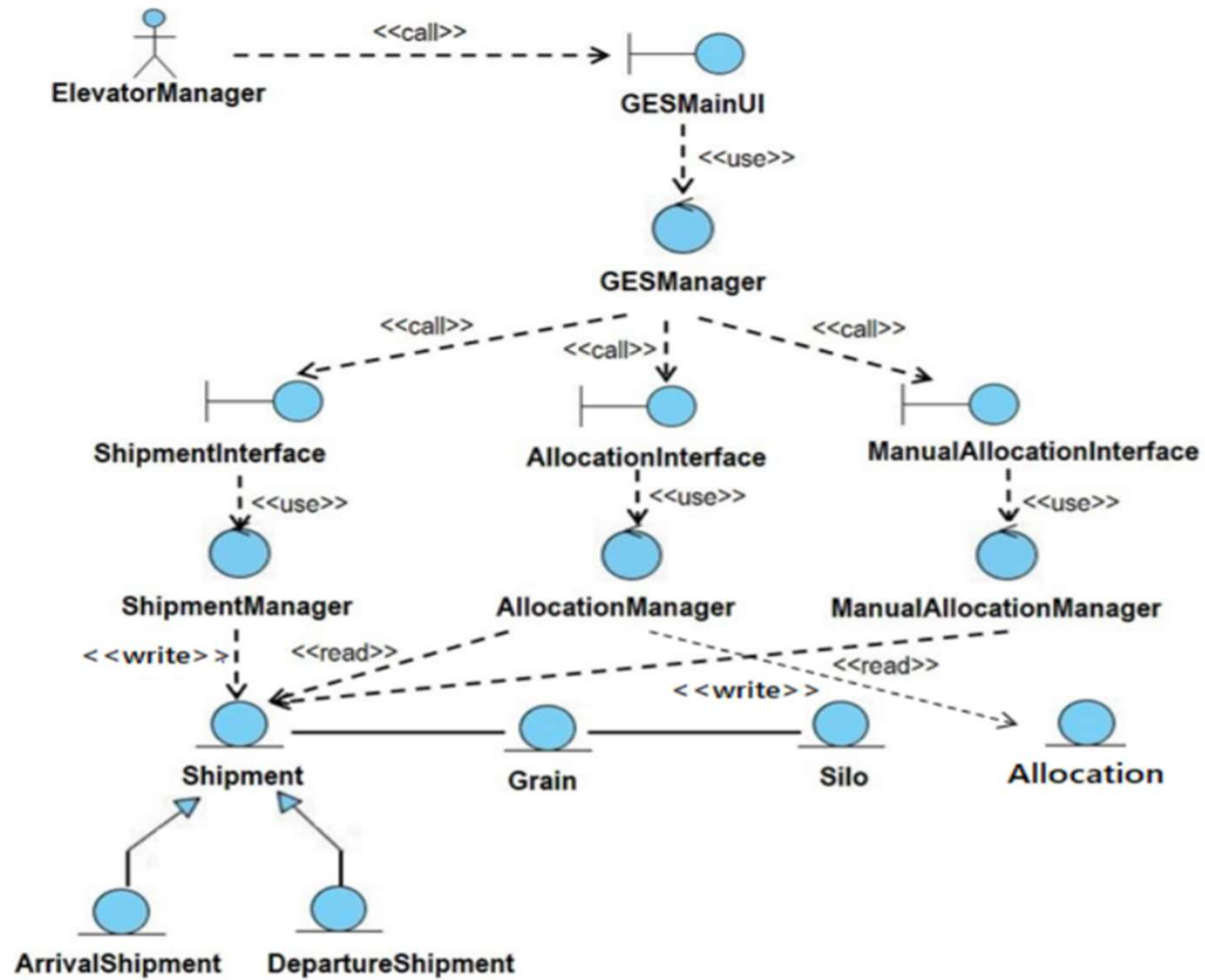
- Distribute the behavior of the use case
 - Assign responsibilities to each object in the form of a set of operations
- Help identify **new** participating objects and **missing** behavior
- Heuristics (layout)
 - The **1st** column: the **actor** who initiated the use case
 - The **2nd** column: a **boundary** object (that the actor used to initiate the use case)
 - The **3rd** column: a **control** object that manages the rest of the use case
- Heuristics (access)
 - Entity objects are accessed by control objects
 - Entity objects never access boundary or control objects







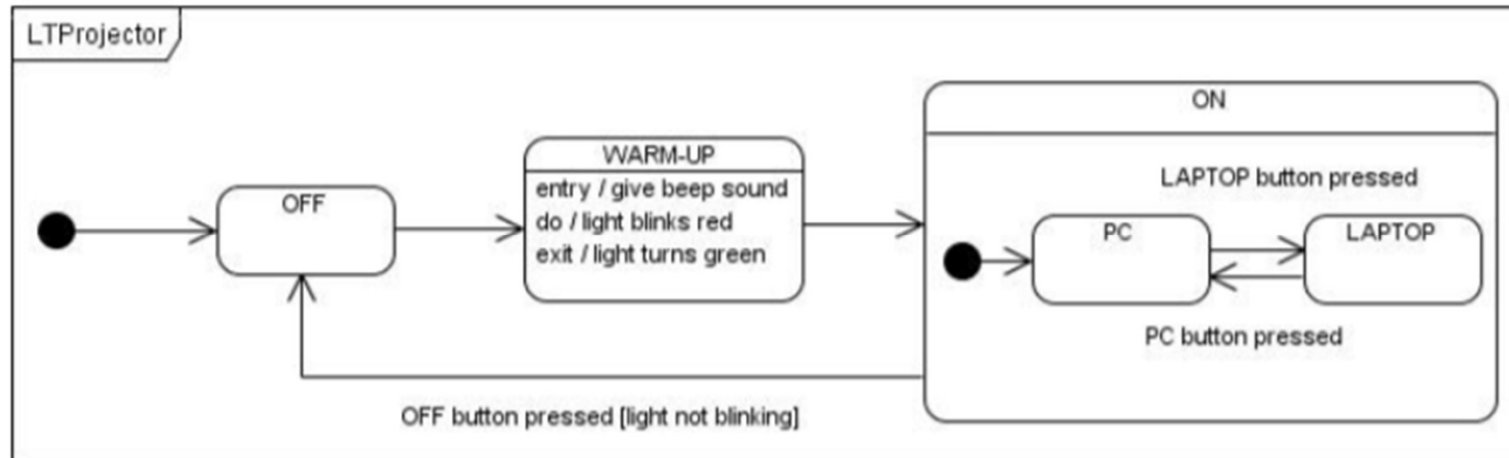
(b) Conceptual Model



Tutorial 5: Requirements Analysis – Dynamic Models

Question 2: Projector Controller Panel

Model the projector's system behavior using a state machine diagram (dialog map).



An alternate state machine diagram

